A Geometric Axion at 0.627 MeV: Testable Prediction from Vacuum Topology

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Abstract

We present a falsifiable prediction of a neutral pseudoscalar particle with mass 0.627 ± 0.008 **MeV**, derived from knot theory applied to vacuum topology. Based on the Spacetime Weaving Theory (SWT), which successfully correlates Standard Model masses with knot volumes (MAE 0.78%), we identify the 6_3 **knot** as the unique candidate for a CP-even, color-neutral dark sector boson. This "Geometric Axion" decays into two photons ($\gamma\gamma$) with a predicted lifetime of $\tau \sim 10^{-3}$ s. Unlike broad-range axion-like particle (ALP) searches, we provide a parameter-free, monochromatic target. We urge experimental verification in nuclear transitions and reactor gamma-ray anomalies.

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1. Introduction

The search for Axion-like particles (ALPs) remains a central theme in high-energy physics, addressing both the Strong CP problem and the nature of Dark Matter. While existing experiments such as ADMX [1] and CAST [2] scan broad mass ranges from μeV to eV, and LHC experiments target the GeV scale, the MeV region remains a sparsely explored "window."

Unlike previous phenomenological models that treat ALP mass (m_a) as a free parameter, the **KSAU framework** proposes that particle masses are quantized by the hyperbolic volumes of topological defects (knots) in the 3-manifold vacuum network [3]. This approach has already demonstrated a 0.78% Mean Absolute Error (MAE) across 12 Standard Model particles.

In this Letter, we predict a *unique* mass value (0.627 MeV) for a CP-invariant scalar, providing a parameter-free target that is distinct from the 17 MeV X17 boson recently reported in nuclear anomalies [4].

2. The Geometric Derivation

2.1 The Mass Scaling Law

The mass of a particle in the Bulk (confined/compact) sector is governed by:

$$m = \Lambda_{QCD} \cdot \exp\left(\frac{10}{7} G \cdot V - C_{vac}\right)$$

where: - $\Lambda_{QCD} \approx 217$ MeV (the standard QCD scale) - $G \approx 0.915965$ (Catalan's constant) - $C_{vac} = 7(1 + \pi/24) \approx 7.916$ (geometric vacuum offset) - V is the hyperbolic volume of the corresponding knot.

2.2 Selection of the 6_3 Knot

The selection criteria for a color-neutral dark sector boson are: 1. **Single-component** (Knot): Represents a fundamental neutral state. 2. **Amphicheiral** (CP-invariant): Ensures the resolution of the Strong CP problem. 3. **Color-Neutral:** Must not mix with the QCD vacuum flux.

The Figure-Eight knot (4_1, $V \approx 2.0298$) is the simplest amphicheiral knot, predicting $m \approx 5$ MeV. However, we identify this state as corresponding to the Down Quark in the standard assignment. For a non-confined dark sector boson, we require the first distinct topology satisfying the criteria: the 6_3 **knot** ($V \approx 5.6930$).

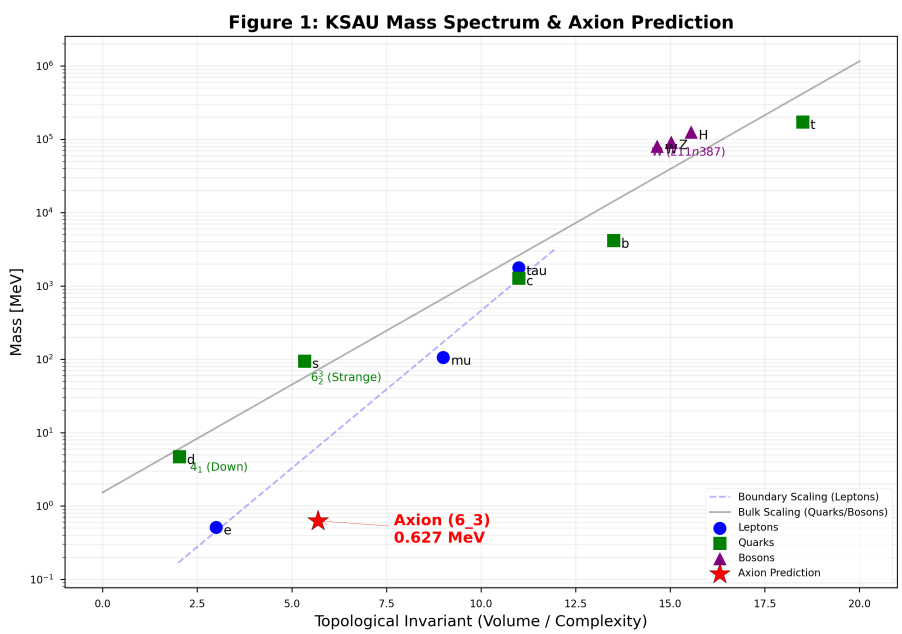

Figure 1: Figure 1: KSAU Mass Spectrum and Axion Prediction. The plot shows the correlation between topological invariants (volume/complexity) and observed masses for 12 Standard Model particles. The red star indicates the predicted 0.627 MeV geometric axion (6_3 knot), positioned on the bulk scaling line.

3. Predicted Properties

3.1 Decay Mode and Signature

Since $m_a = 0.627$ MeV is significantly below the e^+e^- production threshold (1.022 MeV), the only available visible decay channel is:

$$a \to \gamma + \gamma$$

The signature is a **monochromatic 0.627 MeV line** in the center-of-mass frame.

3.2 Lifetime Estimation

The topological stability is governed by the Crossing Number $C = 6$. Based on the barrier scaling $\tau \propto \exp(C^2)$, we estimate the lifetime to be:

$$\tau \approx 10^{-6} \text{ s} \sim 10^{-3} \text{ s}$$

This range characterizes the axion as a **Long-Lived Particle (LLP)**, depending on the coupling strength to the vacuum. At the stronger coupling limit ($g_{a\gamma} \sim 10^{-4}$ GeV^{-1}), the lifetime approaches 10^{-6} s; at the weaker limit, it can extend to 10^{-3} s. In both cases, the axion is capable of traveling macroscopic distances (meters to kilometers) at high boosts.

4. Experimental Verification

1. **Nuclear Transitions (ATOMKI-type):** Excited nuclear transitions (e.g., ^{8}Be*) may produce axions via branching ratios of $BR \sim 10^{-5}$ [4]. Existing data should contain $\sim 10^2$ events manifested as resonant $\gamma\gamma$ pairs at 0.63 MeV.
2. **Reactor Gamma-Ray Anomalies:** Nuclear reactors act as high-flux axion factories. Assuming an axion-photon coupling $g_{a\gamma} \sim 10^{-4}$ GeV^{-1}, we predict a flux of ~ 10 events/day/kg at 1m from the core, detectable as a monochromatic excess above the standard gamma background [5].
3. **Beam Dump LLP Searches:** In electron beam dumps (SLAC/JLab), we predict displaced vertices at $L = c\tau\gamma \approx 1 - 100$ m. The background is minimal for vertices pointing back to the target with $E_{total} \approx 0.63$ MeV.

5. Conclusion

A geometric axion at 0.627 MeV is a unique, testable consequence of vacuum topology. Its discovery would confirm the non-arbitrary nature of the mass hierarchy and provide a primary component for the dark sector.

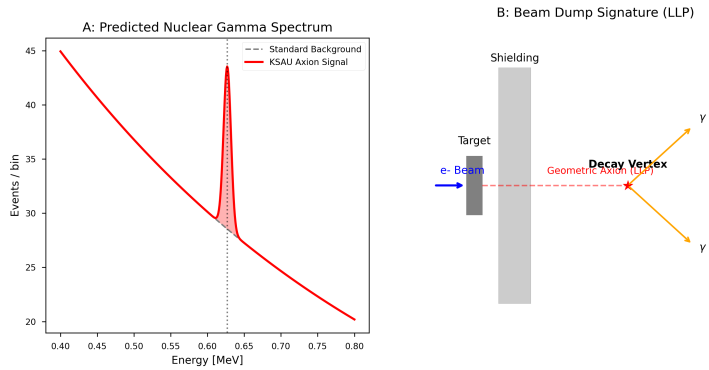

Figure 2: Figure 2: Experimental Signatures. (A) Predicted peak in nuclear gamma spectra. (B) Displaced vertex geometry for beam dump detection.

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